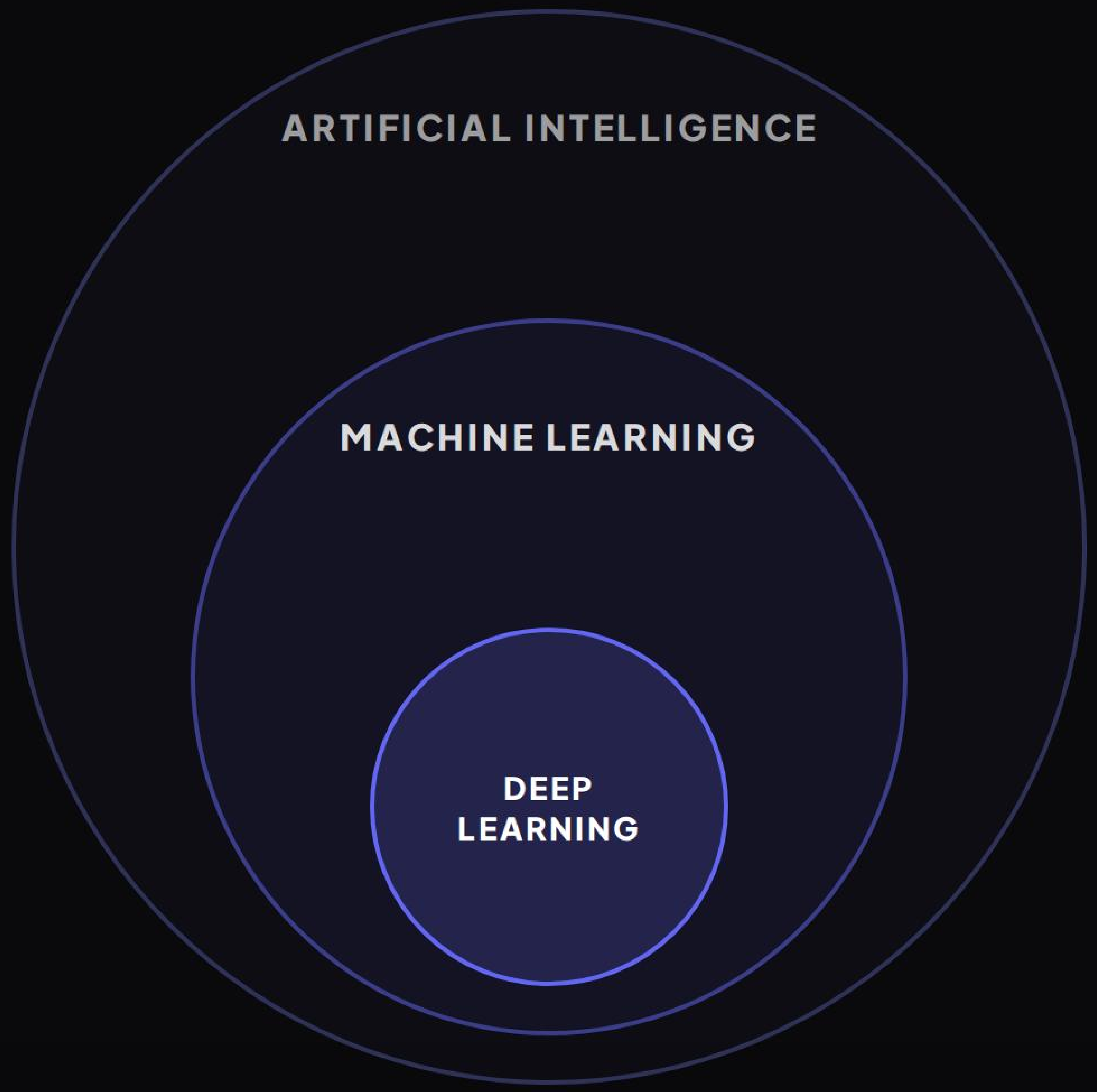
FOUNDATIONS OF AI

Machine Learning Basics

A Comprehensive Visual Introduction

THE LANDSCAPE OF AI



CONVENTIONAL CODING VS. ML

Conventional Coding

- Rules (Hard-Coded Logic)
- Input Data

↓

- Output Results

VS

Machine Learning

- Input Data
- Historical Outputs

↓

- Learned Rules (Model)

THREE PILLARS OF ML

Supervised

The model learns from fully **labeled datasets** containing both input features and the corresponding target answers.

Unsupervised

The model explores **unlabeled datasets** to discover hidden structures, groupings, and underlying patterns independently.

Reinforcement

The system learns optimal behaviors in an environment using continuous feedback loops of **rewards and penalties**.

ML IN THE REAL WORLD

Spam Filtering

Supervised algorithms classify incoming emails based on historical tag mappings from users marking spam.

Customer Segments

Unsupervised clustering groups users by spending habits, allowing targeted promotional campaigns without manual labeling.

Recommendation Engines

Streaming systems predict content choices based on a mix of supervised regression, user ratings, and feedback histories.

SUPERVISED DATA (LABELED)

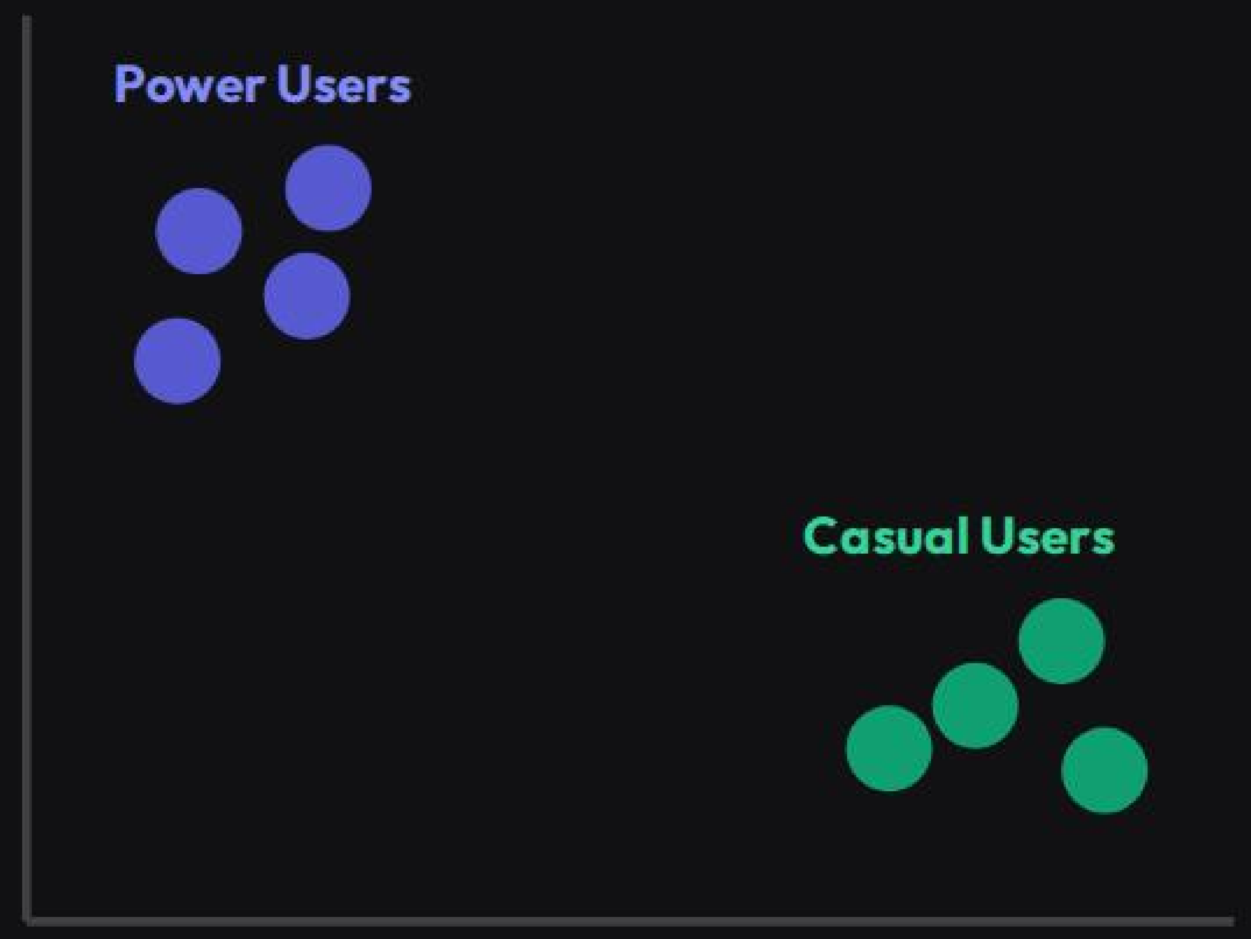
Supervised learning utilizes datasets containing both **Input Features** and a predefined, known **Target Label**.

SENDER DOMAIN	SUBJECT CHARACTERS	CONTAINS "FREE"	EXCLAMATIONS	LABEL (TARGET)
promo-deals@spam.net	32	Yes	5	**SPAM**
manager@workplace.com	14	No	0	**NOT SPAM**
noreply@bankverify.com	24	No	1	**NOT SPAM**
win-cash-today@lottery.org	45	Yes	14	**SPAM**
family@homemail.com	8	No	0	**NOT SPAM**

UNSUPERVISED DATA & CLUSTERING

Unlike supervised learning, there are **no target labels**. The model finds natural groupings directly.

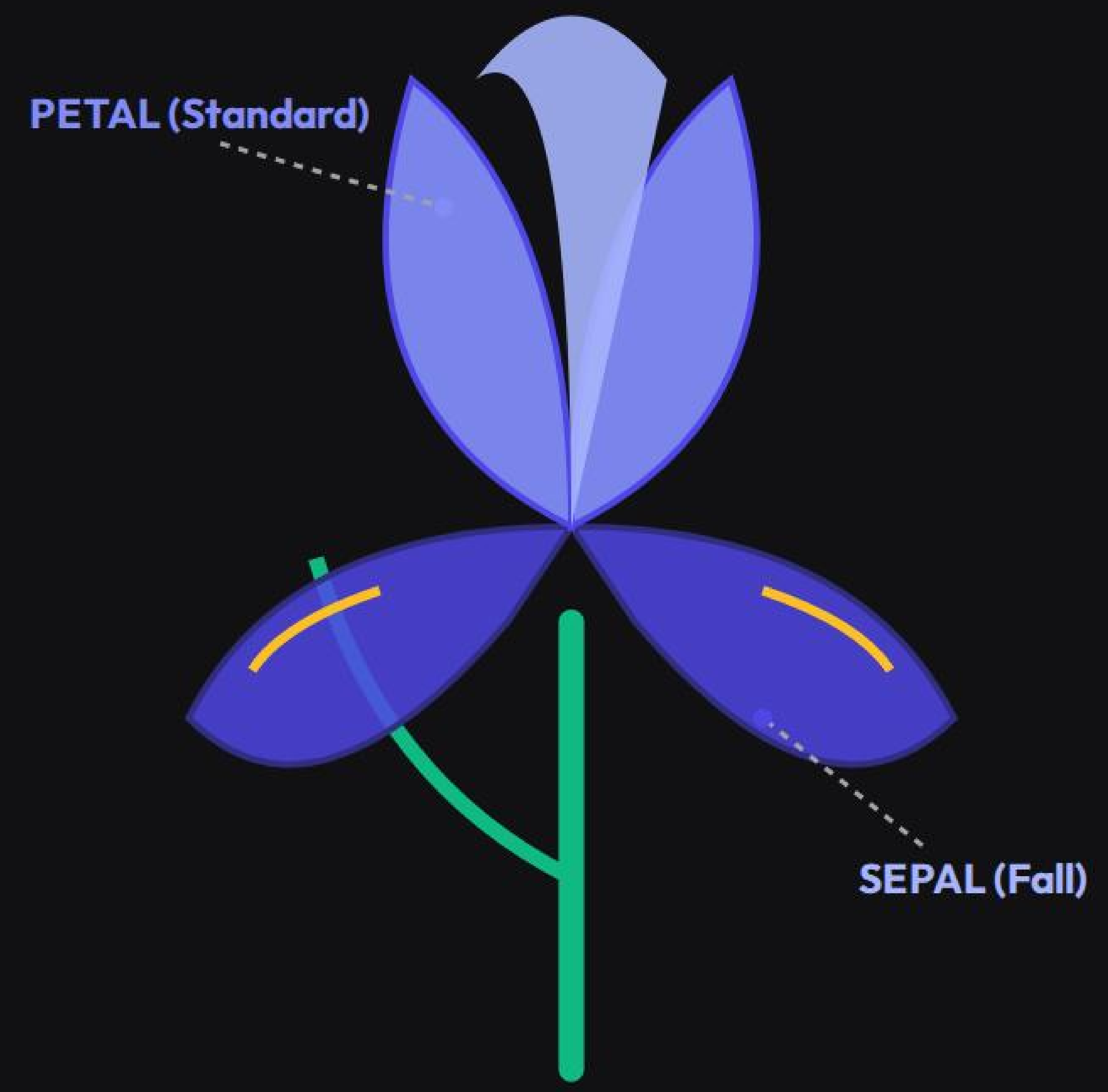
USER ID	AVG. SPEND ($)	LOGINS / WEEK	SESSION MIN
USR_911	850.00	14	38
USR_204	45.00	2	6
USR_733	920.00	11	42
USR_445	60.00	3	8
USR_109	320.00	7	18



THE MACHINE LEARNING PIPELINE

1. PROBLEM DEFINITION
2. DATA FETCHING
3. DATA PRE-PROCESSING
4. EDA (INSIGHTS)
5. FEATURE ENGINEERING
6. MODEL TRAINING
7. MODEL EVALUATION

IRIS CLASSIFICATION ANATOMY



Petal Ratios

The inner upright petals are highly predictive features. Setosa petals are remarkably tiny, while Virginica has the longest petals.

Sepal Support

The lower outer sepals provide foundational width and length measurements used to separate Versicolor from Virginica species.

Three Target Classes

The classic dataset aims to classify three species: **Setosa**, **Versicolor**, and **Virginica**.